MILITARY SPECIFICATION

ENCLOSURE, STANDARD AVIONICS, FORCED AIR COOLED, GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Air System Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the design, manufacture, and testing of forced-air-cooled standard avionics enclosures, hereinafter referred to as enclosures. The enclosures accommodate Standard Electronic Modules (SEM) Format B Span 2 and Format C, and a standard backplane. The enclosures provide cooling for the enclosed devices, protection from the environment, and shielding from Electromagnetic Interference (EMI) for various military avionics systems on multiple platforms.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N3887

FSC 5975

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SPECIFICATIONS

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MIL-B-5087	-Bonding, Electrical, and Lightning Protection, for Aerospace Systems.
MIL-E-5400	-Electronic Equipment, Aerospace, General Specification for.
MIL-T-18303	-Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for.
MIL-T-23103	-Thermal Performance Evaluation, Airborne Electronic Equipment and Systems, General Requirement for.
MIL-C-38999	-Connector, Electrical, Circular, Miniature, High Density, Quick Disconnect, Bayonet, Threaded, and Breach Coupling, Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for.
DOD-C-83527	-Connectors, Plug and Receptacle, Electrical, Rectangular Multiple Insert Type, Rack To Panel, Environment Resisting, General Specification for.

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-Marking for Shipment and Storage.
MIL-STD-454	-Standard General Requirement for Electronic Equipment.
MIL-STD-462	-Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-794	-Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-810	-Environmental Test Methods.
MIL-STD-889	-Dissimilar Metal.
MIL-STD-1389	-Design Requirements for Standard Electronic Modules.

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DOD-STD-1788 -Avionics Interface Design Standard.

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.2 General requirements. The enclosures specified herein shall meet the requirements of MIL-E-5400, for Class 2X equipment.

3.2.1 Proprietary processes. The final design of the enclosure shall not depend on any process, innovation, or feature which would restrain normal multiple source competitive procurement by the government and its contractors.

3.3 Mechanical design.

3.3.1 Size.

3.3.1.1 Enclosure. The external size dimensions, excluding cooling interfaces, of the enclosure shall be in accordance with DOD-STD-1788. The enclosure shall come in two, four, six and eight MCU sizes.

3.3.1.2 Protrusions. All protrusions such as hold-downs, carrying handles, switches, knobs, connectors, and indicators shall lie within an outline envelope in accordance with DOD-STD-1788 in the latched and actuated position.

3.3.2 Modularity. The family of enclosures are to be part of a modular packaging approach which will provide electrical, mechanical, and thermal interfaces for the Standard Avionics Modules. This technique shall provide for the joining of two and four MCU sized enclosures to form four, six or eight MCU sizes with a minimum of additional components. Each enclosure shall accept not less than 26 modules at a 0.30 inch pitch. The modules to be used are the Standard Electronic Modules (SEM) Format B Span 2 Module and the SEM Format C as described in MIL-STD-1389. Table I provides possible combinations of enclosure sizes, module size and backplane module interconnection methods in which enclosures shall be configurable. The maximum enclosure mass value provided refers to the maximum weight of an avionics systems which can be packaged using an enclosure.

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TABLE I. Standard enclosure size and configuration.

Enclosure size	Module size	Backplane interconnect medium	Max. enclosure mass (Kilograms)
2 MCU	SEM Format B	Multilayer board	6.5
4 MCU	SEM Format C	3 Level wire wrap or multilayer board	11.5
4 MCU	2 levels of SEM Format B	2 Multilayer boards	11.5
6 MCU	1 level of SEM Format B 1 level of SEM Format C	2 backplanes using 3 levels of wire wrap each or 2 multilayer boards	16.0
8 MCU	2 levels of SEM Format C	2 backplanes using 3 levels of wire wrap each or 2 multilayer boards	20.0

3.3.3 Backplane interface. The mechanical interface of the backplanes used to interconnect the modules shall be in accordance with figure 1. This applies to both multilayer and wire wrap.

3.3.4 Hold-downs.

3.3.4.1 Front. The enclosure shall have two (one on the two MCU size) NAS622 Type T hooks, or functional equivalents, on the front of the enclosure as shown on figure 2. These hooks shall be used to apply the force required to insert the rear mounted connector and secure the front of the enclosure.

3.3.4.2 Rear. The rear hold-downs of the enclosure shall be either dagger pin receptacles in accordance with figure 3 or a rack and panel connector in accordance with DOD-C-83527 but not both.

3.3.5 Module types. The four MCU enclosure shall be capable of accepting a random mix of SEM Format C modules mounted at pitches of 0.3, 0.4, 0.5, and 0.6 pitch and in increments of 0.3 inch over 0.6 inches. The two MCU enclosure shall accept SEM Format B modules mounted on pitches that are multiples of 0.3 inches. Card guides uniquely machined for a given system are permissible with the four MCU enclosure.

3.3.6 Drain holes. The enclosure shall contain drain holes to allow the escape of fluids in any orientation in order to prevent collection of fluid inside the module compartment.

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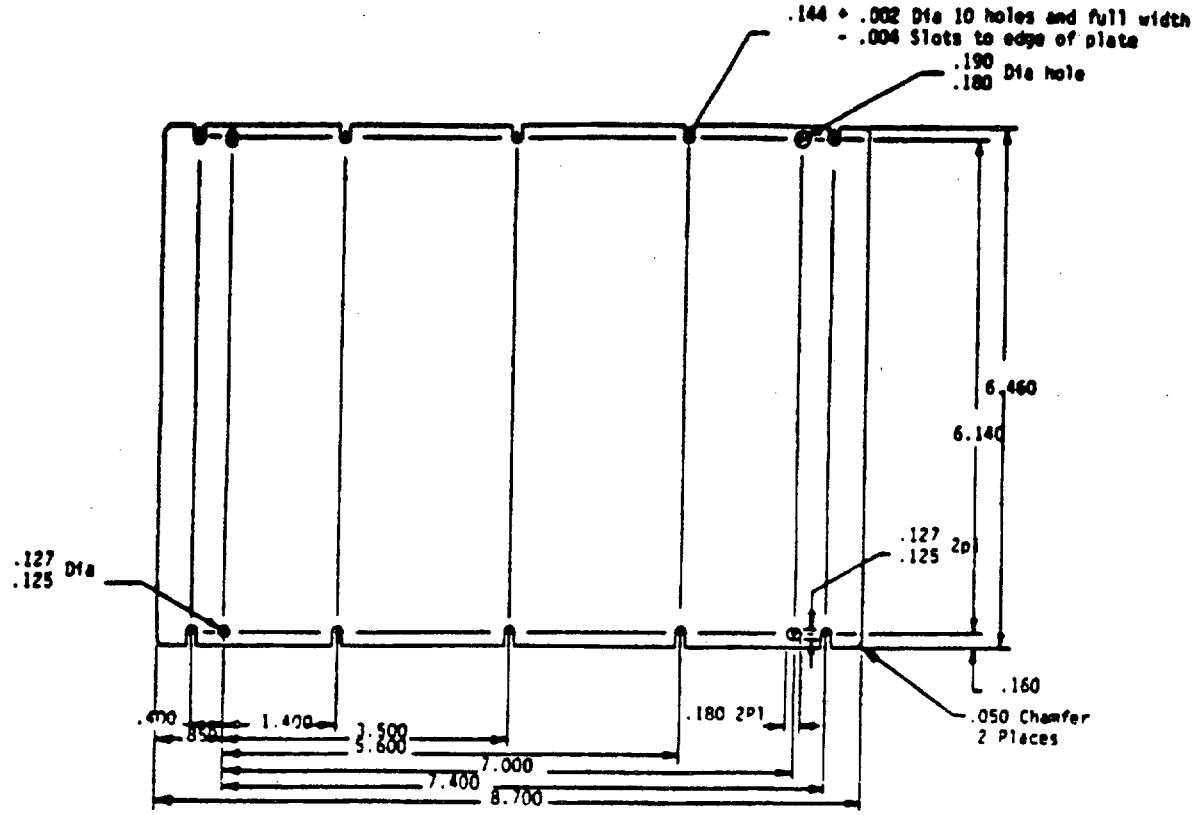
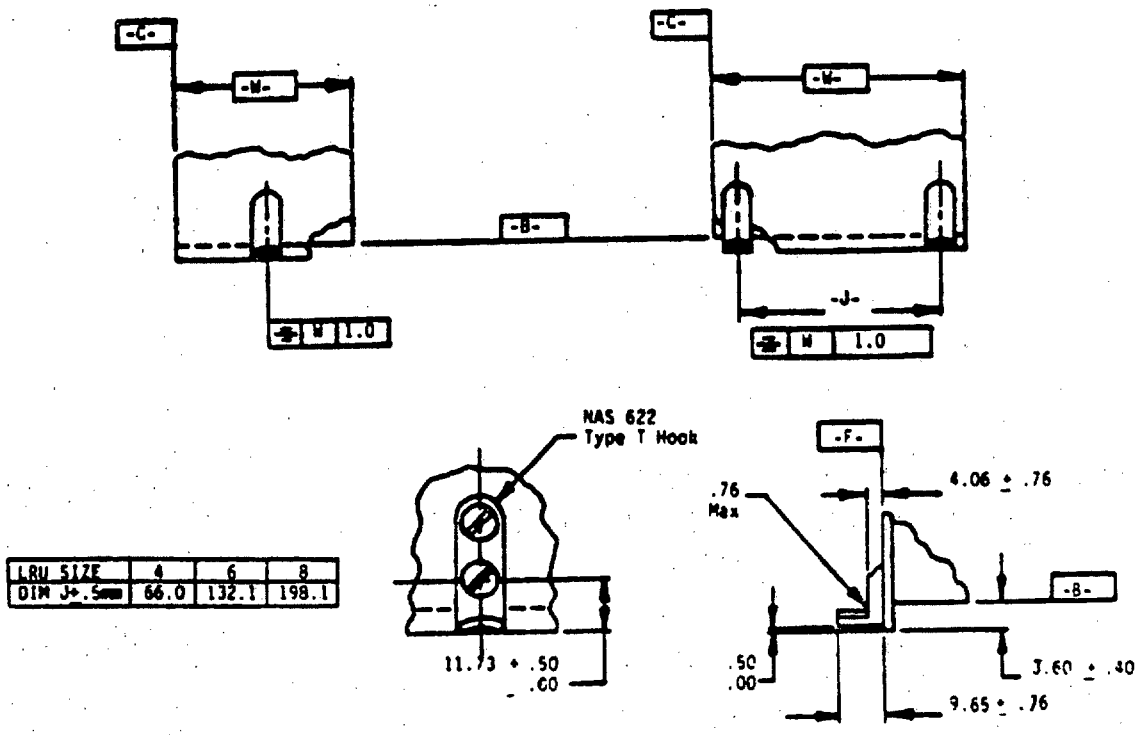


FIGURE 1. Backplane mechanical interfaces.

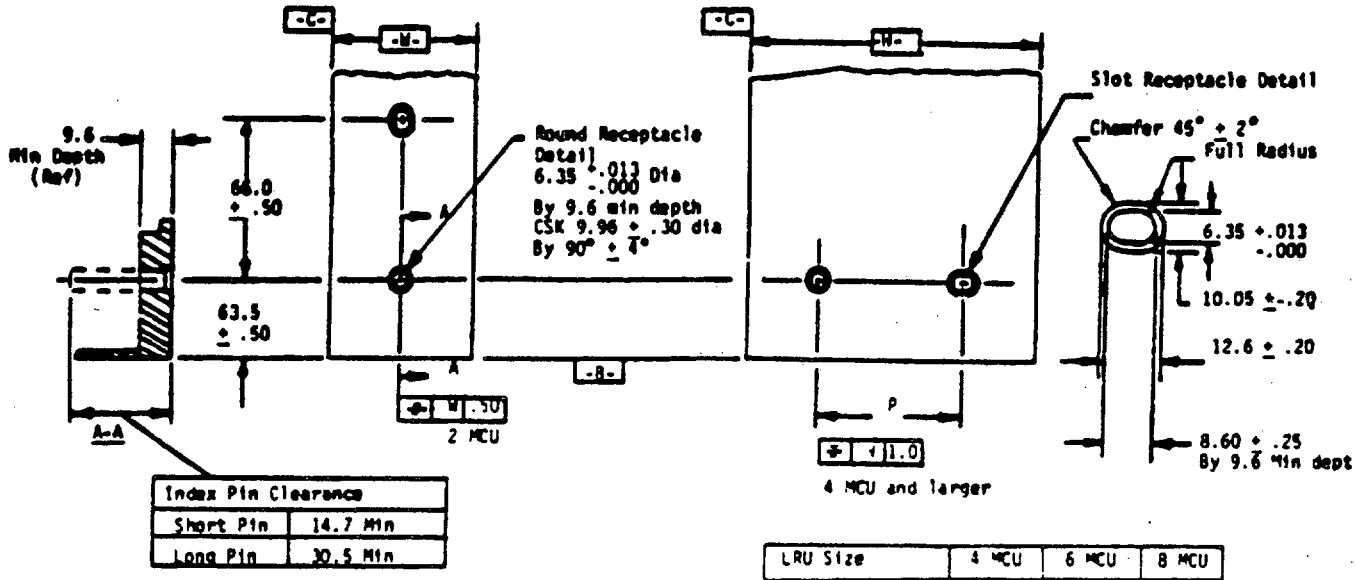
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LRU SIZE	4	6	8
DIM J±.5mm	66.0	132.1	198.1

FIGURE 2. Front hold down.

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Index Pin Clearance	
Short Pin	14.7 Min
Long Pin	30.5 Min

LRU Size	4 MCU	6 MCU	8 MCU
DIM P ± .50 mm	66.0	132.1	198.1

FIGURE 3. Dagger pin receptacles.

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3.3.7 Fluid entrapment. Any feature which may trap a fluid on the exterior of the enclosure, thereby increasing the possibility of fluid entering the module compartment, is not permissible.

3.3.8 Module retainer. The module retainer mechanism shall be part of the enclosure and not part of the module. The module retainer mechanism shall apply an average clamping pressure of 25 +/- 1 PSI on the module. This mechanism shall be shop and depot replaceable.

3.4 Electrical design.

3.4.1 Wiring harness. The enclosure shall be designed to facilitate the removal and installation of the enclosure input/output connector, wiring, EMI filters, and the connector tying these items to the backplane as an assembly.

3.4.2 Meters. The enclosure shall be capable of mounting both a time elapsed meter and built in test (BIT) status indicators. The time elapsed meter chosen shall be military qualified. Electrochemical deposition meters shall not be used.

3.4.3 Input/output connectors.

3.4.3.1 Front. The front mounted connectors shall be in accordance with MIL-C-38999, Series III or IV. Connectors shall be removable from the inside of the enclosure.

3.4.3.2 Rear. The rear mounted rack and panel connector and the position of it in the enclosure shall be in accordance with DOD-C-83527.

3.4.3.3 Connector restriction. Front and rear connectors shall not be used simultaneously.

3.5 Materials.

3.5.1 Standard parts. The enclosure shall use standard (MS, AN, MIL-STD, etc.) parts, materials, and processes wherever feasible. Unless otherwise specified in the contract, a parts control program shall be implemented (see 6.2.1).

3.5.2 Corrosion resistance. Materials shall be of corrosion resistant types or shall be processed to resist corrosion in accordance with MIL-STD-889.

3.5.3 Nameplate. There shall be provisions to mount a nameplate on the front of the enclosure.

3.6 Environment.

3.6.1 High temperature. The enclosure shall not bind, crack, craze, discolor or bulge and the gaskets shall not lose their resiliency, or become permanently distorted when tested as specified in 4.5.1.2.

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3.6.2 Low temperature. The enclosure shall not bind, crack, craze, discolor or bulge and the gaskets shall not lose their resiliency, or become permanently distorted when tested as specified in 4.5.1.3.

3.6.3 Temperature shock. The enclosure shall withstand the temperature shock test as specified in 4.5.1.4 with no physical or performance degradations.

3.6.4 Humidity. The enclosure shall be humidity tested as specified in 4.5.1.5 with no resulting evidence of swelling, absorption of water, change of mechanical properties, corrosion or other forms of deterioration. There shall be minimal entrapped water inside the enclosure.

3.6.5 Fungus. The enclosure shall withstand the fungus test as specified in 4.5.1.6 with no physical or performance degradations.

3.6.6 Salt fog. The enclosure shall be salt fog tested as specified in 4.5.1.7 with no resulting evidence of corrosion, deterioration, or performance degradation.

3.6.7 Acceleration. The enclosure shall be acceleration tested as specified in 4.5.1.8 with no resulting material yield or failure and no loss of joint integrity.

3.6.8 Vibration. The enclosure shall be vibration tested as specified in 4.5.1.9 with no resulting material yield or failure and no loss of joint integrity.

3.6.9 Shock. The enclosure shall be shock tested as specified in 4.5.1.10 with no resulting material yield or failure and no loss of joint integrity.

3.6.10 Crash safety. The enclosure shall be crash safety tested as specified in 4.5.1.11 with no resulting failure of the mounting attachments, however, bending and distortion are permitted. The test item shall remain in place and shall not create a hazard.

3.7 Thermal.

3.7.1 Thermal performance. The enclosure shall be thermal tested as specified in 4.5.2 to show compliance with the minimum thermal performance requirements.

3.7.1.1 Thermal requirement. When tested at the required test conditions, no module guide rib temperature shall be greater than 85 C and cooling air exhaust temperature shall be not greater than 71 C. The airflow rate divided by the enclosure power shall be 1.36 Kg/KW-minute maximum.

3.7.2 Configuration. The enclosure shall be forced-air cooled with air entering the rear of the enclosure and exhausting from the front.

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3.7.3 Leakage. The maximum allowable leakage rate shall not be greater than 0.1 percent of the maximum anticipated flow rate through the enclosure when tested as specified in 4.5.2.4.

3.8 Electromagnetic interference (EMI).

3.8.1 Shielding effectiveness. The enclosure shall be an effective shield against incident electric, magnetic, and electromagnetic field radiation.

3.8.2 Outside-to-inside. The outside-to-inside shielding effectiveness of the standard enclosure shall meet the minimal requirements shown on figure 4 when tested in accordance with 4.5.3.2.

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (4.3).

b. Quality conformance inspection (4.4).

4.3 First article inspection. Unless otherwise specified in the contract, first article tests shall be performed on two units representative of the equipment to be supplied under the contract (see 6.2.1 and 6.2.2). Tests shall be performed in accordance with the approved test procedures.

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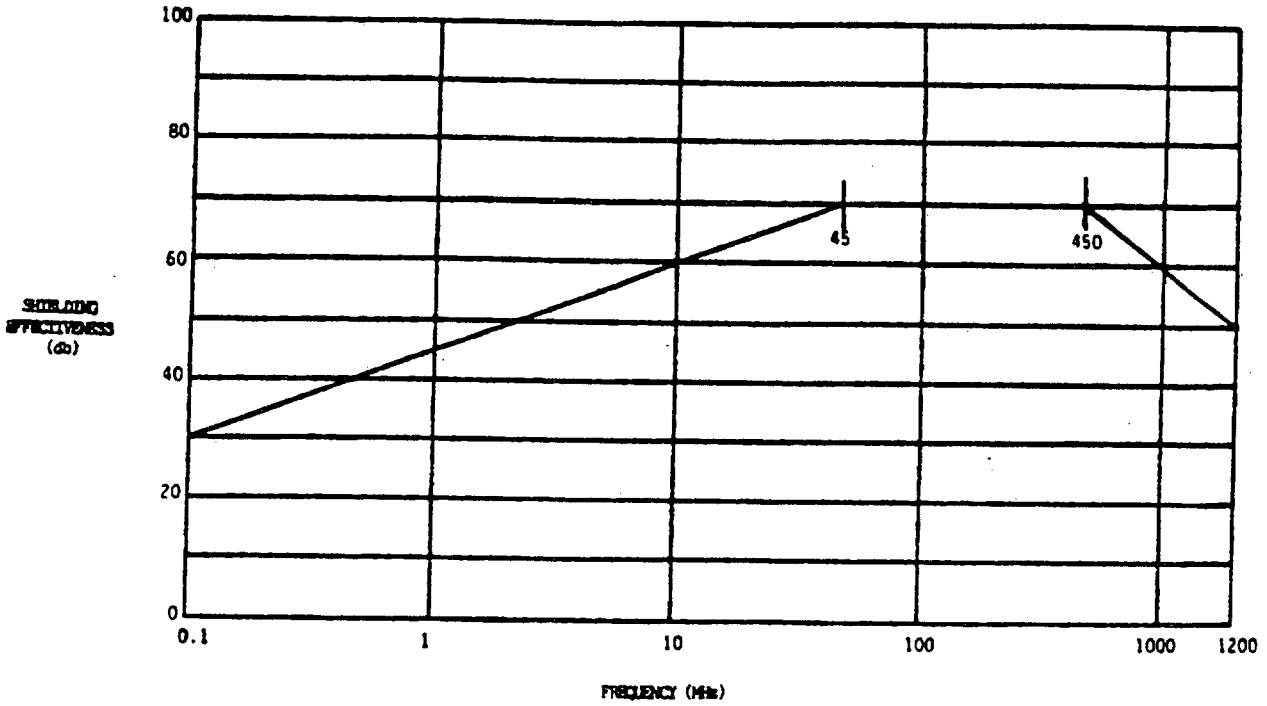


FIGURE 4. EMI shielding design requirements.

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4.3.1 First article inspection routine. First article inspection shall consist of the examinations and tests specified in table II.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Method paragraph
High Temperature	3.6.1	4.5.1.2
Low Temperature	3.6.2	4.5.1.3
Temperature shock	3.6.3	4.5.1.4
Humidity	3.6.4	4.5.1.5
Fungus	3.6.5	4.5.1.6
Salt fog	3.6.6	4.5.1.7
Acceleration	3.6.7	4.5.1.8
Vibration	3.6.8	4.5.1.9
Shock	3.6.9	4.5.1.10
Crash safety	3.6.10	4.5.1.11
Thermal characteristics	3.7.1	4.5.2
Leakage	3.7.3	4.5.2.4
Shielding outside-to-inside	3.8.2	4.5.3.2

4.4 Quality conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.4.1.1 Inspection lot. The inspection lot shall consist of all enclosures of the same size and design produced under essentially the same conditions and offered for inspection at one time.

4.4.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table III.

4.4.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table III.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL percent defective
Visual and workmanship	3.3	--	0%

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4.4.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table IV, in the order shown, and shall be performed on sample units which have been subjected to and have passed the group A inspection.

4.4.1.3.1 Sampling plan. The sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be specified in table IV. Major and minor defects shall be as defined in MIL-STD-105.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL percent defective
Dimensional and mechanical	3.3	--	5%

4.4.2 Periodic inspection. Periodic inspection shall consist of group C inspection. Where the results of these inspections show noncompliance with the applicable requirements (see 4.4.2.1.3), the delivery of products that have passed groups A and B shall not be delayed pending the results of these periodic inspections.

4.4.2.1 Group C inspection. Group C inspections shall consist of the inspections specified in table II, in the order shown. Group C inspection shall be performed on sample units which have been subjected to and have passed groups A and B inspections.

4.4.2.1.1 Sampling plan. Nine units of the same design shall be subjected to group C inspection. The sample shall not be subjected to any screening test prior to submission.

4.4.2.1.2 Failures. If one or more units fail to pass the group C inspection, the sample shall be considered to have failed.

4.4.2.1.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection at the option of the qualifying activity). Groups A and B may be reinstituted, however, final acceptance and shipment shall be

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withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, the manufacturer shall notify the cognizant activity and qualifying activity (see 6.2.2).

4.5 Inspection conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient (25 +/- 10 C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90% relative humidity

4.5.1 Environmental testing.

4.5.1.1 General. The shock, vibration, crash safety, and acceleration tests should be performed with the enclosures containing dummy modules of equal mass which brings the enclosure mass to the maximum (+/- 0.25 Kilograms) specified for that enclosure in table I. The enclosure under test shall only be restrained by its hold-downs.

4.5.1.2 High temperature. The enclosure shall withstand the high temperature test in accordance with MIL-STD-810, Method 501.2, Procedure II for a constant temperature exposure at not less than 95 C.

4.5.1.3 Low temperature. The enclosure shall withstand the low temperature test in accordance with MIL-STD-810, Method 502.2, Procedure II. The test temperature shall not be greater than -57 C.

4.5.1.4 Temperature shock. The enclosure shall be temperature shock tested in accordance with MIL-STD-810, Method 503.2, Procedure I except that the diurnal cycle shall be replaced by a constant temperature. The five minute maximum transfer time and the test duration of one hour or until stabilization whichever is longer shall apply to both hot and cold shocks. Test temperature extremes shall be -57 C and 95 C.

4.5.1.5 Humidity. The humidity test shall be performed in accordance with MIL-STD-810, Method 507.2, Procedure III. This test shall be performed with the enclosure resting on each of three mutually perpendicular surfaces.

4.5.1.6 Fungus. The fungus test shall be performed in accordance with MIL-STD-810, Method 508.3.

4.5.1.7 Salt fog. The salt fog test shall be performed in accordance with MIL-STD-810, Method 509.2, Procedure I.

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4.5.1.8 Acceleration. Acceleration testing shall be performed in accordance with MIL-STD-810, Method 513.3, Procedure I. Acceleration values shall be computed using Table 513.3-I for aircraft with a forward acceleration value of 4g. Test time shall be one minute.

4.5.1.9 Vibration. Testing shall be performed in accordance with MIL-STD-810, Method 514.3, Procedure I. The random vibration envelope shall be in accordance with Figure 514.3-26 Random Vibration Envelope and a Wo of 0.06 G**2/Hz for the functional tests and 0.2 G**2/Hz for the endurance tests.

4.5.1.10 Shock. The test shall be performed in accordance with MIL-STD-810, method 516.3, Procedure I. Test level and duration shall be in accordance with I-3.3.c.(1)(b) for functional test of flight vehicle equipment.

4.5.1.11 Crash safety. The test shall be performed in accordance with MIL-STD-810, Method 513.3, Procedure I. Acceleration values shall be 40g fore and aft, 14g lateral and 20g up and down. Test time shall be one minute. Orientation is in accordance with figure 5.

4.5.2 Thermal testing. The contractor shall plan for thermal testing (see 6.2.2). The tests shall assess the thermal dissipation properties of the enclosure in the following manner using the procedures of MIL-T-23103 except as noted in 4.5.2.1, 4.5.2.2, 4.5.2.3 and 4.5.2.4.

4.5.2.1 Test set-up. The enclosure shall be constrained using its front and rear hold-downs only, with cooling air being supplied through the same hardware as in its final installation. Each enclosure will be fully populated with 26 dummy modules each dissipating an equal amount of power. The dummy module guide ribs shall have dimensions, materials, and plating in accordance with MIL-STD-1389.

4.5.2.2 Test conditions. The test shall be run in a 71 C ambient temperature. Cooling air inlet temperature and pressure drop across the enclosure shall be 27 C and 2.5 inches of water respectively. Input power shall be in accordance with table V.

TABLE V. Standard Enclosure Power Capacity.

Enclosure Size (MCU)	Power Capability (watts)
2	250
4	500
6	750
8	1000

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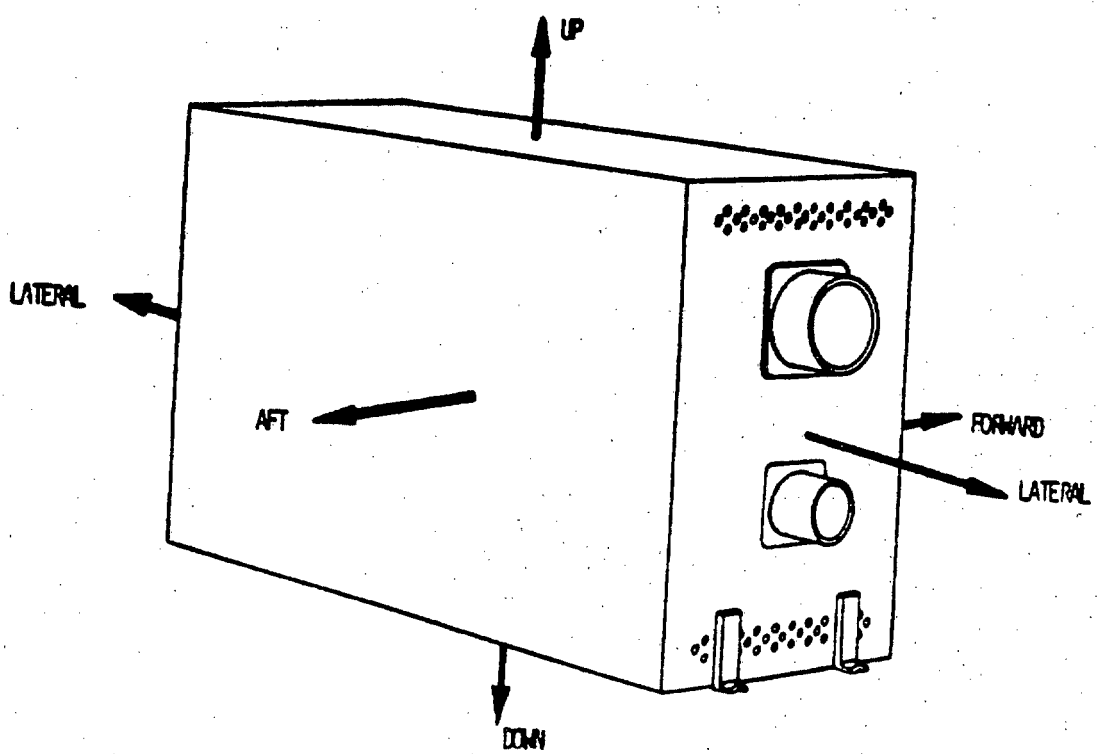


FIGURE 5. Standard enclosure test orientations.

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4.5.2.3 Data. The enclosure shall be instrumented to gather the following data:

a. Inlet air temperature and relative humidity.
b. The guide rib temperatures of modules (numbered sequentially from the air inlet end of the enclosure) 1, 4, 7, 10, 13, 17, 20, 23, and 26 are to be recorded.
c. The exhaust air temperature.
d. The air mass flow rate through the enclosure.

4.5.2.4 Leakage. The requirements of 3.7.3 shall be tested in accordance with contractor-formulated test method (see 6.2.2). The enclosure shall be pressurized to not less than two inches of water at a temperature of 21 C.

4.5.3 Electromagnetic interference testing.

4.5.3.1 Shielding effectiveness. These tests shall demonstrate that the shielding effectiveness of the enclosure complies with the requirements specified in 3.8.

4.5.3.2 Outside-to-inside. The requirements of 3.8.2 shall be tested in accordance with contractor-formulated test method (see 6.2.2). The test setup shall be in accordance with MIL-STD-462.

5. PACKAGING

5.1 Standard enclosure preservation and packing. Preservation and packing shall be in accordance with Level A, B or C as specified in MIL-STD-794, as is specified in the contract, without the use of preservation compound (see 6.2.1).

5.2 Packing. Shipping containers may contain a multiple number of enclosures. The container shall be suitable for Level A, B or C as specified in MIL-STD-794.

5.3 Unitized loading. Unitized loading for the enclosure is authorized as specified in MIL-STD-794.

5.4 Marking. All shipping containers shall be marked in accordance with the requirements of MIL-STD-129 with the following information:

a. ENCLOSURE, STANDARD, (mcu size) MCU
b. MILITARY SPECIFICATION NO. MIL-E-85726
c. PRESERVED (date)
d. DOMESTIC OR EXPORT PACKED (as applicable)
e. GOVERNMENT ORDER NO. (or contract no. if order no. is not assigned)
f. NAME OF SUPPLIER (and name of manufacturer if not the same)
g. MANUFACTURER'S CODE IDENTIFICATION NUMBER

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6. NOTES

6.1 Intended use. The enclosure covered by this specification is intended to accommodate functional avionics modules. It is intended for use on piloted aircraft. The user shall identify the electrical, cooling, and mounting structure interfaces, and their appropriate requirements. Also, the format, type, and location of modules shall be identified.

6.2 Ordering data. Acquisition documents shall specify the following items.

6.2.1 Acquisition requirements.

a. Title, number and date of this specification.
b. Implementation of a parts control program, if required (see 3.5.1).
c. Responsibility for inspection (see 4.1).
d. Name and location of government approved test laboratory (see 4.1).
e. Number of equipment subjected to first article inspection, if required (see 4.3).
f. Approval of previously used test procedures, if required (see 4.3).
g. Modification of previously used test procedures, if required (see 4.3).
h. Selection of applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).
i. Mounting configuration and required interfaces.
j. Cooling resources and necessary appropriate interfaces.
k. Number, format, and placement of modules on backplanes.
l. Electrical connector type and placement.

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID no.	Option
(a) 4.3	First Article Inspection Procedures	DI-T-4901	--
(b) 4.3	First Article Inspection Report	DI-T-4902	--
(c) 4.4.2.1.3	Failure/Malfunction Report	UDI-T-23724	--

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(d)	4.5.2	General Test Plan/Procedures	DI-T-3707	--
(e)	4.5.2.4	General Test Plan/Procedure	DI-T-3707	--
(f)	4.5.3.2	General Test Plan/Procedure	DI-T-3707	--

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 Data requirement waivers. The data requirements of 6.2.2 may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Quality assurance definitions. Definitions for quality assurance terms are in accordance with MIL-STD-109.

6.4 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of the first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. If the enclosure is procured in conjunction with a specific functional system, the system modules should be inserted into the enclosure and tested as a complete system. The contracting officer must be specific in regard to the manner in which first article inspection is to be conducted.

6.5 Subject term (key word) listing.

Aircraft
Avionics
Backplane
Enclosure
Modules

Preparing activity:
Navy (AS)

(Project 5975-N-)

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6.5 Subject term (key word) listing.

Aircraft
Avionics
Backplane
Enclosure
Modules

Preparing activity:
Navy (AS)

(Project 5975-N-647)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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